

Vivek Kumar Ray

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Experience

Software Developer, Google Summer of Code(CircuitVerse) – remote, India May 2025 – Aug 2025

- Contributed to Circuitverse, and got selected for google summer of code internship with more than 30+ PRs merged.
- Worked on the feature that allows conversion of Circuits to Verilog and vice-versa and took the feature to production. Integrated the Yosys server for CircuitVerse

Linux Kernel Developer, Vicharak.in – Surat, India Aug 2025 – Present

- Worked on Board bring-up,Linux Driver & Controller Debugging, Camera stack(Qualcomm and Rockchip) and User level Operating System software application

Software Developer Intern, FOSSEE(eSim) – mumbai, India Nov 2024 – May 2025

- Repackaged Circuit Simulator eSim 2.4 desktop application with new features for the launch of 2.5 version
- Created the eSim tool-chain for Ubuntu to handle and manage its resources and other improvements to the application

Co-founder and CTO, GoBuild.in – Jammu, India Aug 2023 – Dec 2024

- Developed a multi model & multi data pipeline agentic construction management software or (CMS) and developed CRM. Designed the UI for the official website and created the frontend, backend, APP(Deployed on PlayStore), Ai SAAS tools,and database for the company

Projects

GoBuild CMS [Link](#) [Code](#) [Demo](#)

- Developed a agentic SAAS application for GoBuild. It is a multi-model and multi data pipeline application that allows you to use various agents for fault detection, resource management, data intelligence, weather management and others.
- Tools Used: TS,Nextjs,LangGraph,LLaMa, OpenAi, Pinecone, Prisma, GCP cloud run.

Copilot for Circuit Design and Simulation [Link](#)

- Developed a copilot-like feature in CircuitVerse's platform to build circuit automatically, using LLMs. It allows the users to describe the circuit in natural language, which then converts it into circuit using verilog and allow you to simulate it using CircuitVerse's platform.
- Tools Used: Pinecone, VueJS, Yosys, CircuitVerse, PostgreSQL, OpenAI API, TypeScript, Redis, Shadcn UI

Multi Party computation protocol using Oblivious Trasfer Algorithm [Link](#)

- Created a communication protocol on top of TCP/IP using the Correlated-Oblivious Transfer Algorithm. It is used to perform multi-party computation without sharing secrets between the participants.
- Tools Used: C++ , boost-asio, TypeScript, trezor-crypto, Protobuf.js, nanopb, nodeJS, @noble/curves/secp256k1

Li Battery SOH & RUL Estimation [Link](#)

- Built a project to predict State of health (SoH) for identifying remaining useful life of Li-ion batteries.
- Tools Used: Python,Sklearn, Keras , matplotlib, numpy, Tensorflow, seaborn

Linux Device Driver [Link](#)

- A Char Device driver for linux kernel, it consists of a dynamic circular queue, that uses IOCTL calls to talk to user programs
- Tools Used: C, Makefile, Linux

Open Source Contributions and Education:

- **Education:** Btech ECE (SMVDU 2022-2026)
- **CircuitVerse:** Raised 20 PRs, with more than 15 successfully merged. Got GSoc 2025 fellowship.
- **eSim (FOSSEE):** Raised 10 PRs, all of which were merged. Got full semester internship